

Summarizing and displaying categorical data

EDUC 641: Unit 1

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Roadmap

<i>Research is a <u>partnership</u> of questions and data</i>		What types of data are collected?	
		Categorical data	Continuous data
What kinds of questions can be asked of those data?	Descriptive questions	- How many members of class have black hair? - What proportion of the class attends full-time?	- How tall are class members, on average - How many hours per week do class members report studying, on average?
	Relational questions	- Are male-identifying students more likely to study part-time? - Are PrevSci PhD students more likely to be female-identifying?	- Do people who say they study for more hours also think they'll finish their doctorate earlier? - Are computer-literate students less anxious about statistics?

This unit borrows heavily (with permission) from John B. Willett's course S-010Y

Class goals

1. Understand and implement principles of tabular (rectangular) data in R
2. Describe and summarize quantitative data that are categorical
3. Create visualizations of quantitative data that are categorical
4. Write R scripts to conduct these analyses

The story behind the data

- [Warren McCleskey](#) sentenced to death for murdering police officer during armed robbery in Georgia (1978)
- McCleskey appeals to the Supreme Court, arguing his death sentence is due to racial bias in sentencing...his appeal is rejected
- Senator Edward Kennedy sponsors the Racial Justice Act to enforce "the Constitution's promise of equality under the law"
- McCleskey executed in Georgia (1991)



- [Baldus, Pulaski & Woodworth \(1983\)](#). A comparative review of death sentences: An empirical study of the Georgia experience. *J. Crim Law*.

Our motivating question: Were convicted murderers more likely to be sentenced to death in Georgia if they killed someone Black or if they killed someone White?

Note: Racial bias may exist in the criminal justice system irrespective of our answer to this question. Exploring such an issue would knit together more complete evidence from multiple research traditions.

Goals of the unit

We'll use a simplified version of the quantitative data accumulated by Baldus et al. (1983) to:

- Understand and implement principles of tabular data in R
- Describe and summarize quantitative data that are categorical
- Create visualizations of quantitative data that are categorical
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Materials

1. Death penalty data (in file called deathpenalty.csv)
2. Codebook describing the contents of said data (deathpenalty_codebook.pdf)
3. R script to conduct the data analytic tasks of the unit (EDUC641_3_code.R)

Let's first access the data

Baldus, Pulaski and Woodworth (1983) compiled criminal justice and crude demographic information on all convicted murderers in the state of Georgia from in the mid-1970s. We have access to a simplified version of these data.

```
d <- read.csv('C:/Users/daviddl/Documents/EDUC/EDUC641_22F/data/deathpen
# but there are a lot of reasons to avoid using a hard-coded filepath
# so instead, we'll use our first R package

# Side note, I can write anything I want after a # sign and it
# doesn't cause any problems for your program. These are called
# comments and are critical to remind you (and tell others) what
# you are doing!
```

Let's first access the data

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```
# The first thing we'll do is to install the package  
# install.packages("here") <- when you do this, remove the #  
# don't forget the quotes around the package name!  
# Then, we'll load the package:
```

```
library(here)
```

While you're at it, do the same for **tidyverse**. The **tidyverse** is a collection of helpful packages for manipulating and visualizing data. In particular, it includes the extremely useful **dplyr** and **ggplot2**!

Let's first access the data

Baldus, Pulaski and Woodworth (1983) compiled criminal justice and crude demographic information on all convicted murderers in the state of Georgia from in the mid-1970s. We have access to a simplified version of these data.

```
# Ok, now we're ready to get started:  
  
df <- read.csv(here("data/deathpenalty.csv"))  
# it is common in R to name our datasets short names  
# so as to reduce typing (df for dataframe)
```

but where did our data go?

Understanding data structure

df

```
#>      id deathpen rdefend rvictim
#>  1     1         0        2        2
#>  2     2         0        2        1
#>  3     3         1        2        2
#>  4     4         0        2        2
#>  5     5         0        1        1
#>  6     6         0        2        1
#>  7     7         0        1        1
#>  8     8         0        2        2
#>  9     9         0        2        2
#> 10    10         0        2        1
#> 11    11         0        1        1
#> 12    12         0        1        1
#> 13    13         0        2        2
#> 14    14         0        1        1
#> 15    15         0        2        2
#> 16    16         0        2        2
#> 17    17         0        1        1
#> 18    18         0        1        2
#> 19    19         0        1        1
#> 20    20         0        2        2
```

Understanding data structure

```
names(df)
```

```
#> [1] "id"          "deathpen" "rdefend"  "rvictim"
```

```
head(df)
```

```
#>   id deathpen rdefend rvictim
#> 1  1         0       2        2
#> 2  2         0       2        1
#> 3  3         1       2        2
#> 4  4         0       2        2
#> 5  5         0       1        1
#> 6  6         0       2        1
```

Some vocabulary:

- **Observation**: each row contains information on one person
- **Variable**: each column (R stores these as "vectors")
- **Values**: entries in the column

Putting words to numbers

```
str(df)
```

```
#> 'data.frame':    2475 obs. of  4 variables:
#> $ id          : int  1 2 3 4 5 6 7 8 9 10 ...
#> $ deathpen: int  0 0 1 0 0 0 0 0 0 0 ...
#> $ rdefend : int  2 2 2 2 1 2 1 2 2 2 ...
#> $ rvictim : int  2 1 2 2 1 1 1 2 2 1 ...
```

We know from our codebook what this long list of numbers means, so let's make our dataset a little more readable...

```
df$deathpen <- factor(df$deathpen,
                      levels = c(0,1), labels = c("No", "Yes"))
head(df)
```

```
#>   id deathpen rdefend rvictim
#> 1  1      No   White   White
#> 2  2      No   White   Black
#> 3  3     Yes   White   White
#> 4  4      No   White   White
#> 5  5      No   Black   Black
#> 6  6      No   White   Black
```

Questions go with data

What sorts of questions could you ask already of this data?

Based on the contents of this data, what questions can you think of that ask you to describe single variables?

Based on the contents of this data, what questions can you think of that ask you to inquire about relationships between two or more variables?

Summarizing data: Tables

One powerful thing to do with data is to just count it up. How many defendants were sentenced to death?

```
table(df$deathpen)
```

```
#>  
#>   No   Yes  
#> 2346  129
```

From the *table* of the values of the variable *DEATHPEN*, we conclude ...

Summarizing data: Charts

Even more powerful is to visualize these counts!

```
counts <- table(df$deathpen)
barplot(counts, xlab = "Sentenced to Death?")
```



Sometimes, *proportions* are better ... can you calculate by hand the proportion of defendants who are sentenced to death?

Summarizing data: Proportions

We can also ask R to do this for us

```
prop <- prop.table(table(df$rvictim))  
barplot(prop, xlab = "Race of Victim",  
        ylab = "Proportion of victims")
```

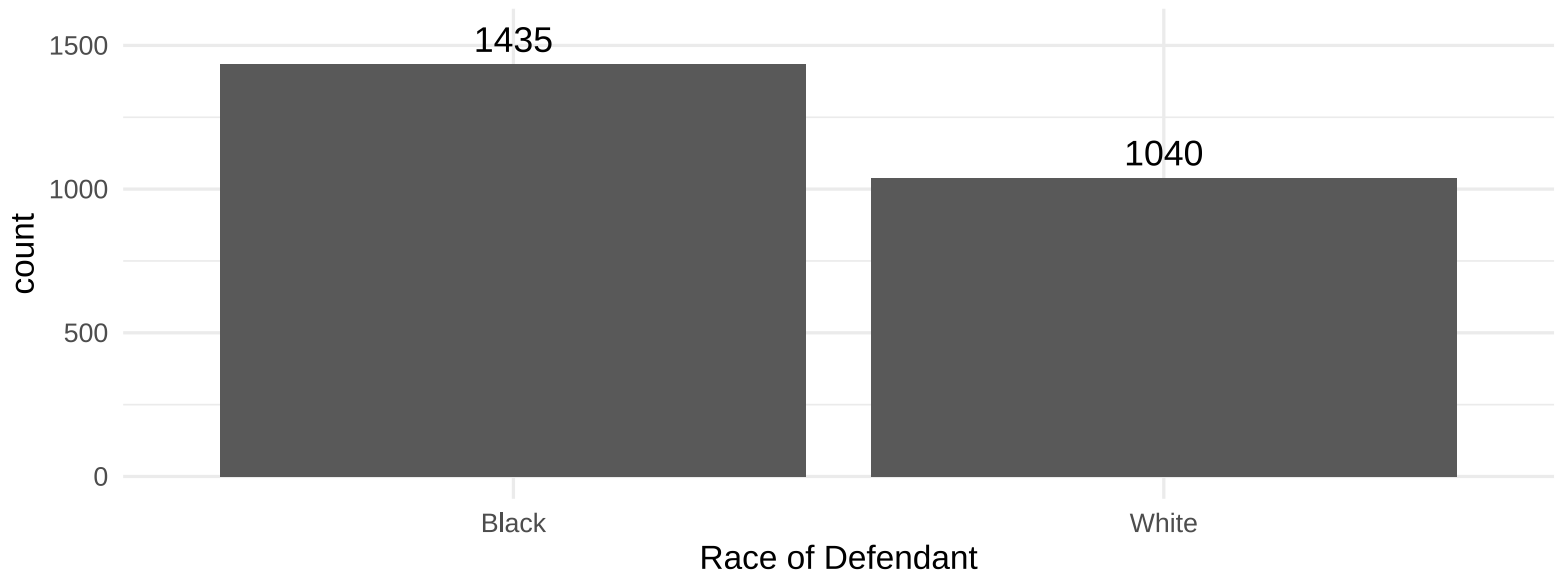


After inspecting this *chart of the proportions* of the variable *RVICTIM* in our data, we conclude ...

Another way to visualize

Let's start to get familiar with the beauty of `ggplot`

```
rd <- ggplot(df, aes(rdefend)) + geom_bar() +  
  xlab("Race of Defendant")
```



After inspecting this *chart* of the values of the variable *RDEFEND* in our data, we conclude ...

BUT...

We still have not answered our motivating question: **Were convicted murderers more likely to be sentenced to death in Georgia if they killed someone Black or if they killed someone White?**

We will do so next class and start our journey into the wild world of **statistical inference!**

(how's that for a cliffhanger?)

Synthesis and wrap-up

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To-Dos

Reading:

- LSWR Chapter 6: an overview of graphing in R; great to return to again and again

Optional follow-up

- Complete R Bootcamp Module 3 (installing and loading a package)
- Complete R Bootcamp Module 9 (importing data)
- Complete R Bootcamp Module 11 (creating plots)
- Complete R Bootcamp Module 12 (creating a project)

Assignment:

- Assignment #1 Due October 12, 11:59pm

Quiz